

VIMAL SELVARAJAN

+1 (510) 598-5492 | vimalselvarajan@gmail.com | LinkedIn | GitHub

EDUCATION

University of California, Riverside

Expected June 2027

B.S. in Computer Science with Business Applications

Riverside, CA

Honors: Dean's List (7 quarters)

Relevant Coursework: Computer Architecture; Operating Systems; VLSI Design; Embedded Systems; Algorithms

TECHNICAL SKILLS

Languages: C/C++, Python, Verilog, Assembly, Bash, Java, SQL

Architecture: Intel Simics, MacSim, Intel XED, gem5, cycle-level simulation, caches, PIM

Tools & Hardware: Linux, Git, CMake, GDB, pytest, Altium, STM32CubeIDE, SPI, I2C, CAN, USB, FTDI/MPSSE

RESEARCH EXPERIENCE

Brisk Lab, University of California, Riverside | Advisor: Prof. Philip Brisk

June 2026–Present

Undergraduate Researcher, Computer Architecture

Riverside, CA

- Build a documented, reproducible Intel Simics–MacSim pipeline in C/C++, Intel XED, and Python for multicore instruction tracing, binary-trace generation, and sampled warm-up and detailed execution
- Design an adaptive cache-warming strategy for MacSim using last-level-cache cold-set tracking and optimistic and pessimistic IPC bounds to reduce simulation warm-up overhead
- Validate traces, configurations, and checkpoints; analyze CPI and IPC across cycle-level experiments

Sadredini Lab, University of California, Riverside | Advisor: Prof. Elaheh Sadredini

April 2026–Present

Undergraduate Researcher, Secure Computer Architecture

Riverside, CA

- Investigate secure PIM for untrusted hardware using arithmetic secret sharing and garbled circuits
- Analyze security and performance trade-offs of off-chip data exposure, integrity risks, and CPU bottlenecks
- Develop workload-partitioning and precomputation strategies for secure PIM on UPMEM-style accelerators

ENGINEERING EXPERIENCE

Hastest Solutions, Inc.

June 2024–August 2024

Software Engineer Intern

San Jose, CA

- Automated a 1,000-hour high-temperature operating-life test of 48 RF amplifier modules by developing and deploying a Python validation platform that collected performance and reliability data
- Brought up the end-to-end test station with customers and hardware engineers, integrating Keysight DAQ970A instrumentation and programmable power supplies

Highlander Racing (Formula SAE), UC Riverside

January 2024–June 2024

Associate Firmware Engineer

Riverside, CA

- Designed multi-layer driver-dashboard PCBs in Altium Designer for the second-generation car, including an STM32F-series interface to an AMOLED display, prioritizing signal integrity
- Developed and debugged STM32F405RGT ADC firmware in STM32CubeIDE over JTAG, improving signal accuracy and reliability, and integrated CAN bus for real-time vehicle data and fault detection
- Produced schematics and manufacturing documentation to align engineering and production teams

PROJECTS

HTOL Test Automation Suite | Python, PyVISA, PyFTDI, pandas

GitHub

- Automated gate-bias tuning across four RF amplifier stages with an iterative `adjust_gate_voltage` algorithm that steps a DAC until drain current reaches target
- Built a five-layer FTDI MPSSE SPI stack for the AMC7836 DAC and a PyVISA library controlling a Keysight DAQ970A and three power supplies, with CSV logging

MTPLite: Read Selection for Long-Read Assembly | Python, Bash, Jellyfish, hifiiasm, QUAST

GitHub

- Assembled *C. elegans* chromosome I into 3 contigs (99.993% genome fraction, 0 misassemblies) from only 14,176 of 1.31M HiFi reads (1.08%) chosen by the MTPLite pipeline
- Implemented k-mer read encoding in a memory-mapped index, lazy greedy set-cover anchor selection, and bounded path search to recover connecting reads

PRESENTATIONS

Selvarajan, V. "Genome Assembly Optimization Using k-mer-Based Read Selection." CRA UR2PhD Poster, April 2026.